

Question	Marking details	Marks Available					
		AO1	AO2	AO3	Total	Maths	Prac
6	<u>Mutation</u> A1 Mutations are {spontaneous / random events / very low in frequency} A2 The change in structure of {gene / chromosome} or change in {sequence of bases / DNA} A3 Are the source of genetic variation A4 ionising radiations / e.g. gamma radiation / UV / X-rays A5 cigarette smoke / smoking / cigarettes / polycyclic hydrocarbons A6 carcinogens / mutagens / chemicals causing cancer / benzenes / asbestos <u>Gene mutation</u> B1 Ref to point mutation / Ref to {deletion / addition / substitution / inversion} of bases / reference to frame shifts / {nonsense / stop / missense} mutations B2 (Gene mutations) can affect protein synthesis / or description of B3 changing the sequence of amino acid / code for a different amino acid B4 Can change the phenotype of an organism B5 Some mutations have no effect on the phenotype / or description of degenerate code / mutation in introns / silent mutations B6 Example of gene mutation: Cystic fibrosis / sickle cell anaemia <u>Chromosome mutation</u> C1 Chromosomal mutations affect many genes / different sequence of genes (on chromosomes) C2 (Different types of chromosome mutations) e.g. {inversion / translocation / deletion} of part of a chromosome / non-disjunction (in meiosis) C3 Most chromosome mutations occur during crossing over (in prophase-I) C4 Could result in {trisomy / aneuploidy / change in chromosome number} C5 may be fatal C6 Example of chromosome mutation: Downs Syndrome	6	3		9		